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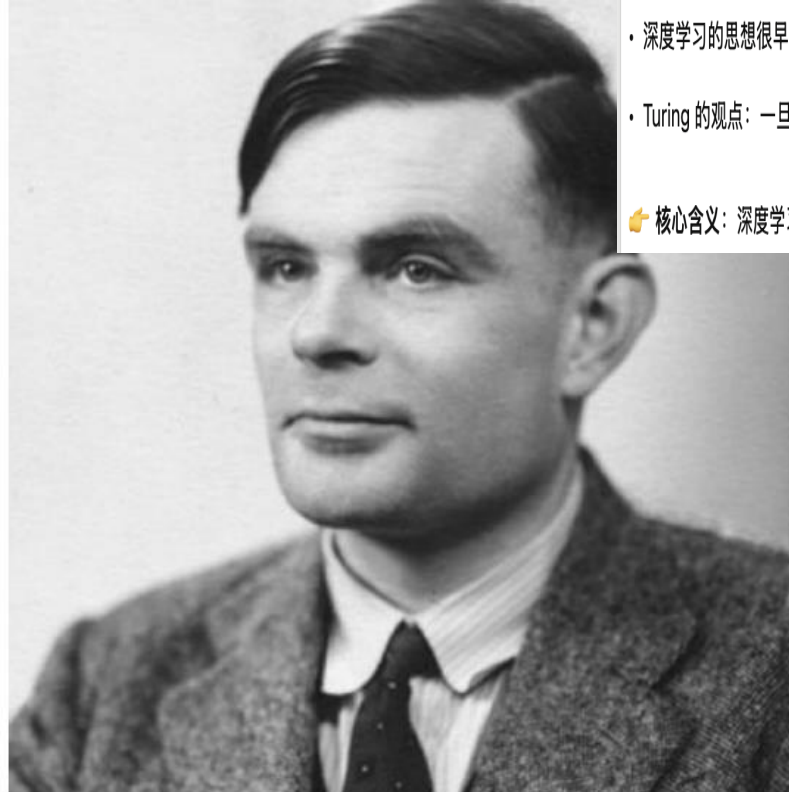
COMP3055

Machine Learning

Topic 15 – Deep Learning Basics

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2025 Autumn

History of Deep Learning Ideas



- 深度学习的思想很早就出现了，最早可追溯到 Alan Turing (1951)。
- Turing 的观点：一旦机器可以模拟人类思维，它们的能力会迅速超过人类。
- 👉 核心含义：深度学习并不是最近才“突然出现”的，而是其数学基础、工程推进逐步成熟才走到今天。

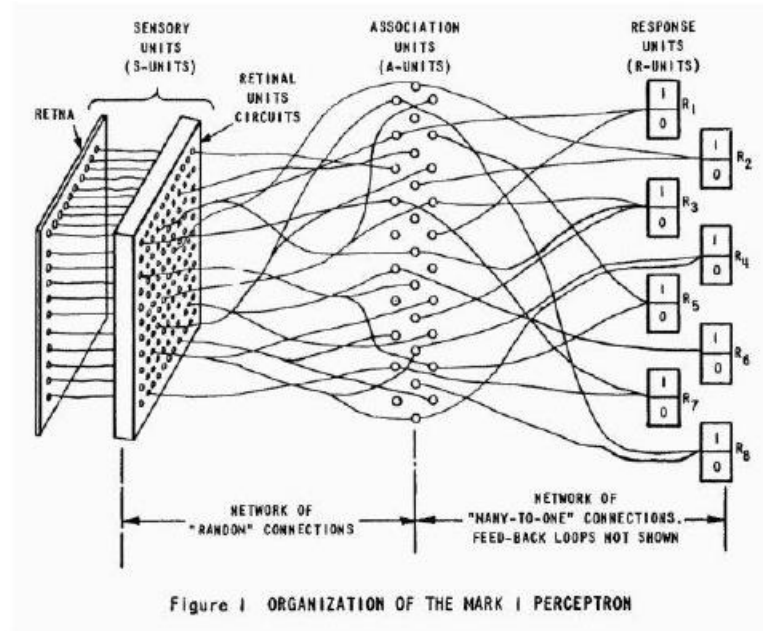
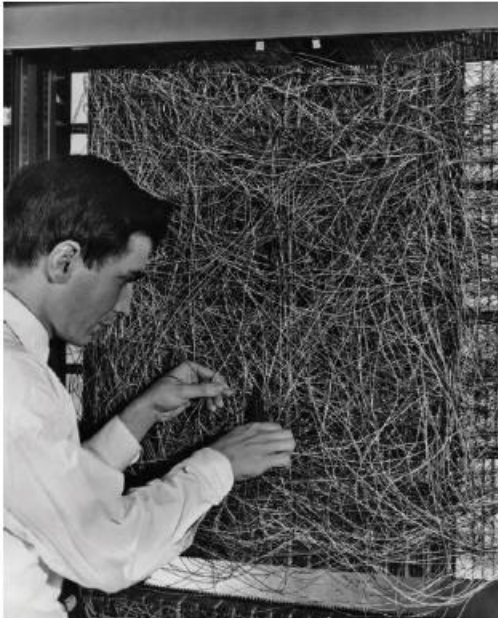
Dreams, mathematical foundations, and engineering in reality.

Alan Turing, 1951: "It seems probable that once the machine thinking method had started, it would not take long to outstrip our feeble powers. They would be able to converse with each other to sharpen their wits. At some stage therefore, we should have to expect the machines to take control."

Milestones

- 1943: Neural networks
- 1957-62: Perceptron
- 1974-86: Backpropagation, RBM, RNN
- 1989-98: CNN, MNIST, LSTM, Bidirectional RNN
- 2006: “Deep Learning”, DBN
- 2009-12: ImageNet + AlexNet
- 2014: GANs
- 2016-17: AlphaGo, AlphaZero
- 2017-19: BERT, Transformers

Milestones



Frank Rosenblatt, Perceptron (1957, 1962): Early description and engineering of single-layer and multi-layer artificial neural networks.

Milestones



Lee Sedol vs AlphaGo, 2016

Turing Award for Deep Learning



- Yann LeCun
- Geoffrey Hinton
- Yoshua Bengio

Turing Award given for:

- “The conceptual and engineering breakthroughs that have made deep neural networks a critical component of computing.”

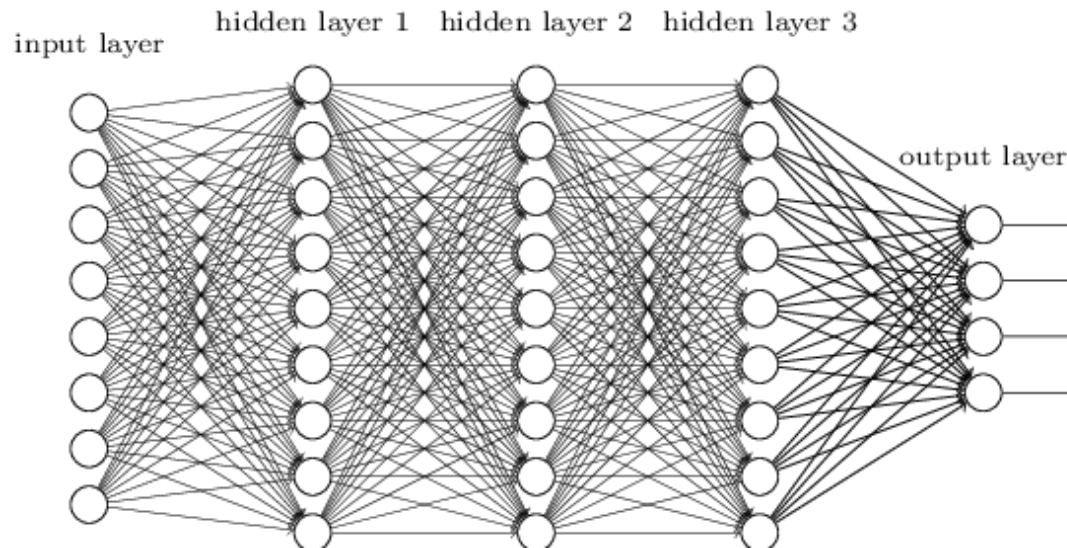
What is Deep Learning?

大多数机器学习方法之所以效果良好，是因为有由人类设计的输入特征或表示形式

- Most machine learning methods work well because of human-designed input features or representations
 - Our job is to find the best features to send to learning techniques such as SVM. 我们的工作就是找到最佳特征，将其发送给诸如支持向量机（SVM）之类的机器学习技术。
- Machine learning becomes just optimizing weights to best make a final prediction. 机器学习就变成了仅仅优化权重以做出最佳最终预测。

What is Deep Learning?

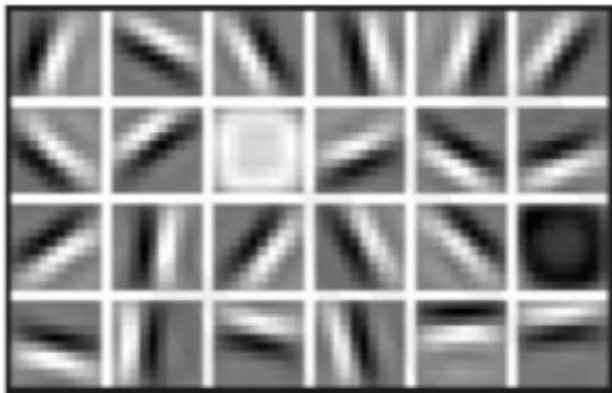
- In contrast to **standard machine learning** 与标准机器学习不同
- **Representation learning** attempts to automatically **learn good features or representations** 表征学习试图自动学习良好的特征或表征
- **Deep learning algorithms** learn **multiple levels of representations** (here: hidden layer 1, 2, 3) and **an output layer** 深度学习算法学习多个层次的表征（此处：隐藏层 1、2、3）和一个输出层
- From “raw” inputs x (e.g. sound, pixels, characters, or words) 从“原始”输入 x （例如声音、像素、字符或单词）
- **Neural networks** are the currently **successful method for deep learning** 神经网络是目前深度学习的成功方法
- A.k.a. “**Differentiable Programming**” 又称“可微编程”



Why Deep Learning?

- To learn the underlying features directly 直接学习底层特征

Low Level Features



Lines & Edges

Mid Level Features



Eyes & Nose & Ears

High Level Features



Facial Structure

Why Deep Learning?

- To learn the underlying features directly

💡 为什么要“深” (Why Deep)

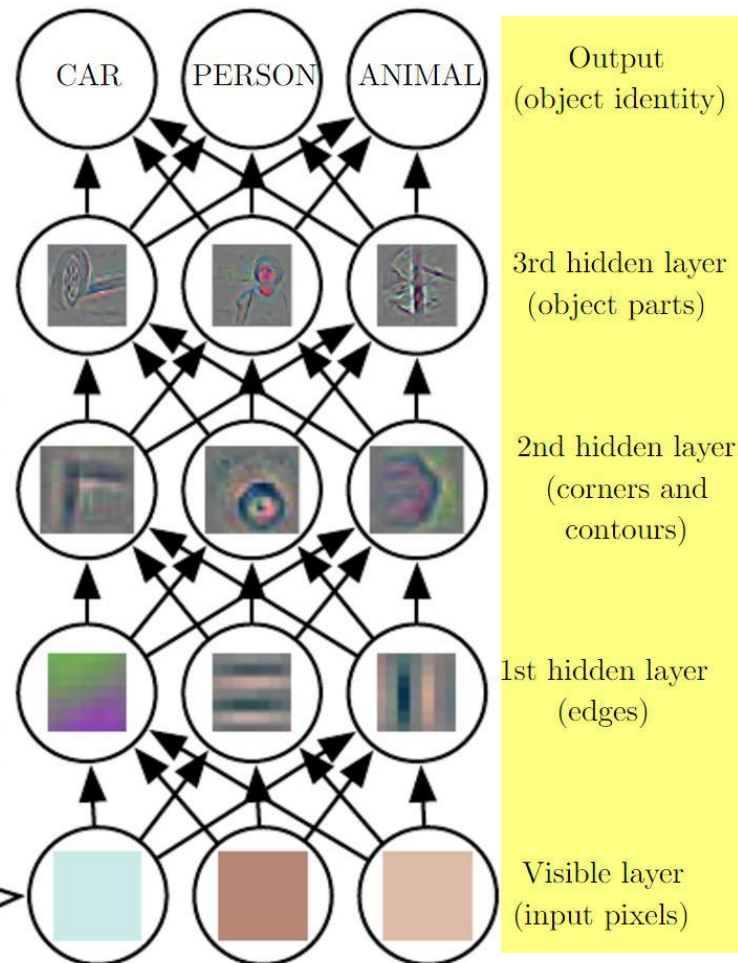
因为:

- 浅层网络只能学到低层特征 (边、线)
- 深度网络可以学习层层递进的抽象表示
- 这些高级抽象表示决定了模型的识别能力

例如:

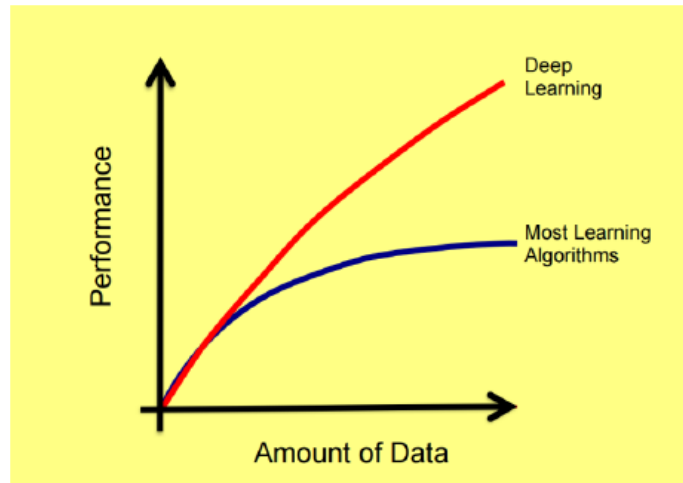
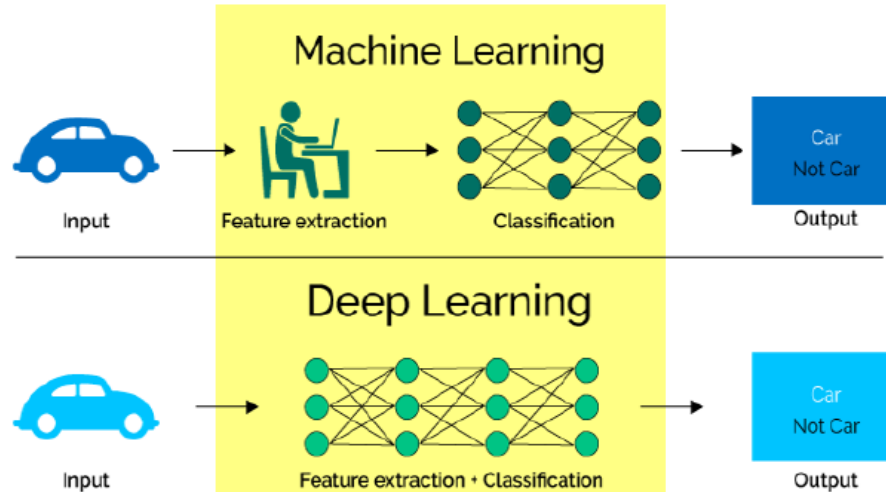
- 人脸识别: 必须学到眼睛鼻子嘴巴的组合
- 车的识别: 必须学到轮子、车灯、车身
- 狗的识别: 必须学到耳朵、鼻子、毛发

👉 深度学习之所以强, 是因为它自动构建了一个“特征金字塔”。



Why Deep Learning?

- Scalable machine learning 可扩展机器学习



Why Now?

- Big data

- Easier collection and storage 采集和存储变得容易

- Larger datasets (ImageNet, COCO)

IMGENET

- Hardware

GPU 原本是为游戏渲染图像设计的，它有一个特点：可以同时做成千上万的并行计算 (massively parallelizable)

- Graphics Processing Units (GPUs)

- Massively parallelizable 并行计算



- Software

- Toolboxes 工具箱

- Powerful framework

要素	解释
Big Data	提供足够训练材料 (ImageNet、COCO 等)
Hardware (GPU)	提供计算能力，否则模型太慢
Software (PyTorch/TF)	降低门槛，让所有人都能训练模型

 TensorFlow  PyTorch

Why Deep

更复杂的模型：深度网络（多层）比浅层网络（1 – 2 层）更有“表达能力”（Representation Power）

- **More sophisticated models**

能学习到更多好的特征（表示）：深度网络（多层）比浅层网络（1 – 2 层）更有“表达能力”（Representation Power）

- Learn more good features (representations)

从而获得更好的分类结果：高级抽象特征 = 更好的分类性能

- So better classification results



Problems with Going Deeper

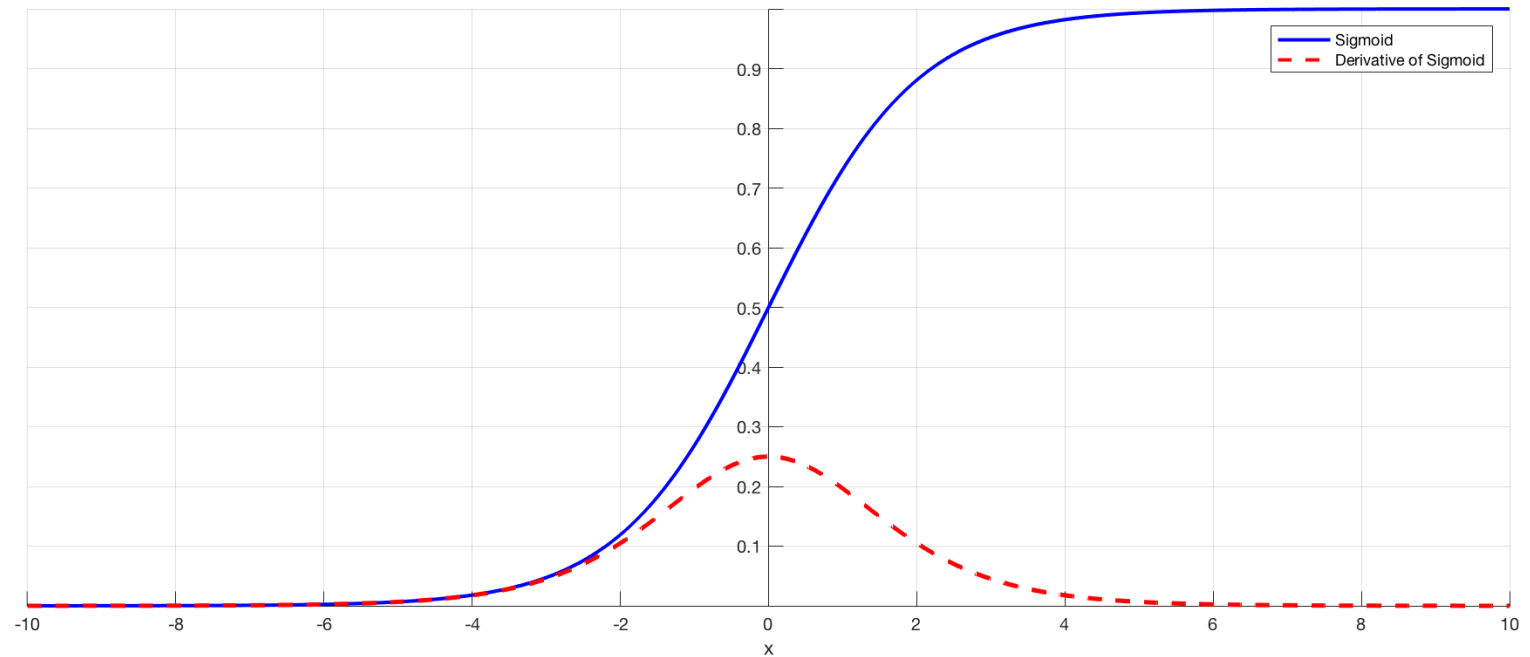
- **Vanishing gradients** 梯度消失

在神经网络中添加更多使用某些激活函数的层时，损失函数的梯度会趋近于零，从而使网络难以训练

- As more layers using certain activation functions are added to neural networks, the gradients of the loss function approaches zero, making the network hard to train
- Certain activation functions, like the sigmoid function, squishes a large input space into a small input space between 0 and 1. Therefore, a large change in the input of the sigmoid function will cause a small change in the output. Hence, the derivative becomes small

某些激活函数，如 S 型函数，会将较大的输入空间压缩到 0 到 1 之间的较小空间。因此，S 型函数输入的大幅变化只会导致输出的小幅变化。因此，导数会变得很小

Vanishing Gradients



Vanishing Gradients

Use better activation function 使用更好的激活函数

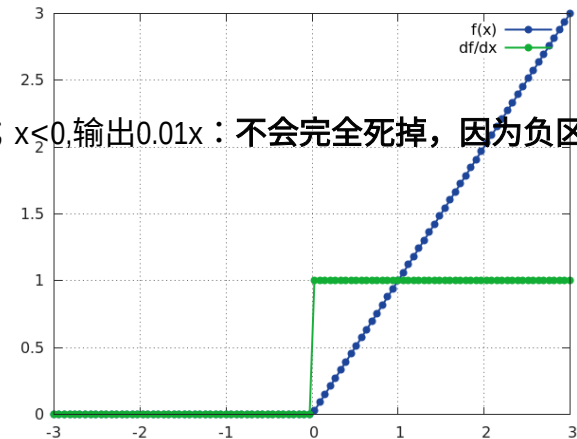
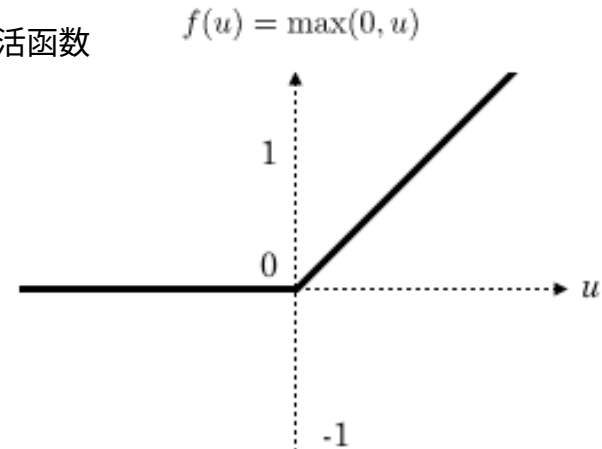
- **Rectified Linear Units (ramp)**

- $f(x) = \max(0, x)$ $x < 0$, 输出 0 ; $x > 0$, 输出 x
- Derivative: **All in or all out (unit step)**
 - $f'(x) = 1$ if $x > 0$ else 0

ReLU 不会把导数压缩成很小的值，因此能有效避免梯度消失。ReLU 的导数（要么 1，要么 0）。

- **Dead ReLUs**

- **LeakyReLU: $f(x) = \max(x, 0.01x)$** $x > 0$, 输出 x ; $x \leq 0$, 输出 $0.01x$: 不会完全死掉，因为负区间仍有梯度流动。
- **PReLU: $f(x) = \max(x, ax)$**

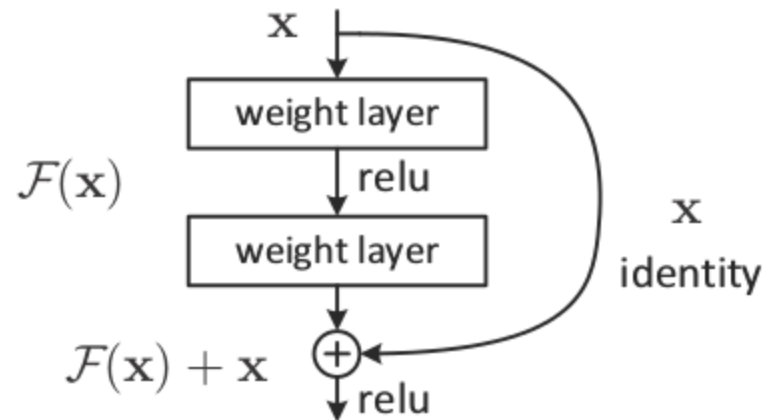


Vanishing Gradients

Use better architecture 使用更好的网络结构

- **Residual networks**

- provide residual connections straight to earlier layers 提供直接连接到早期层的残差连接
- This residual connection doesn't go through activation functions that “squashes” the derivatives, resulting in a higher overall derivative of the block
这种残差连接不经过会“压缩”导数的激活函数，从而使得整个模块的导数更大

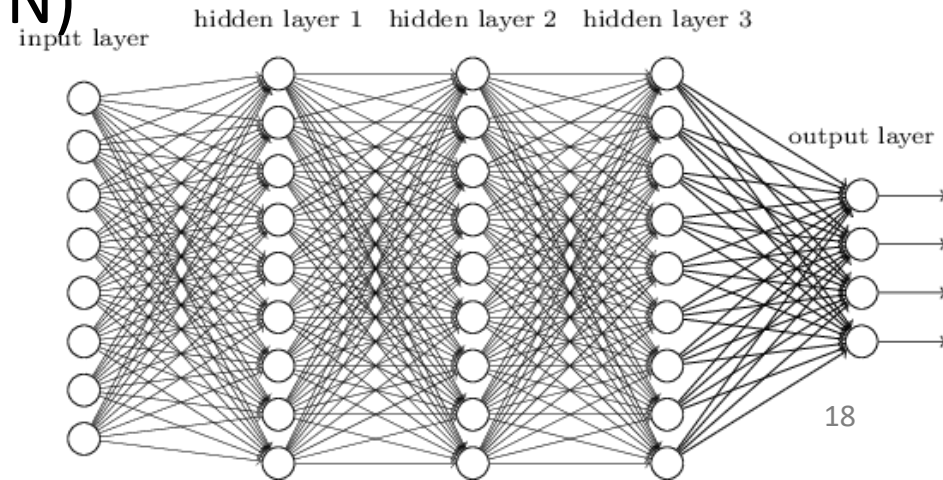


Use normalization 使用归一化

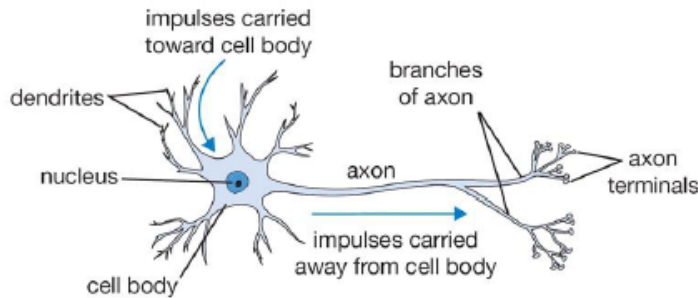
- Batch normalization BN 的作用：规范化每一层的输入分布

Problems with Going Deeper

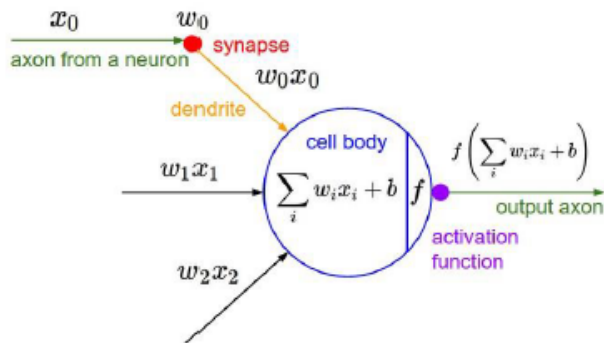
- **Parameter explosion** 参数爆炸：如果堆很多层时，会导致参数爆炸：参数太多导致训练成本高；容易过拟合；优化困难，空间更复杂
 - 随着网络深度增加，需要优化的权重过多
 - **Too many weights to optimize** as we go deeper 搜索空间更难探索：参数空间越大-》优化问题越难
 - Search space is **much harder to navigate**
- **Proposal: shared weights** 解决方案：共享权重
 - Spatially shared (CNN) 空间共享（卷积神经网络）
 - Temporally shared (RNN) 时间共享（循环神经网络）



Biological and Artificial Neural Networks



- **Neuron:** computational building block for the brain



- **(Artificial) Neuron:** computational building block for the “neural network”

Key Difference:

- **Parameters:** Human brains have $\sim 10,000,000$ times synapses than artificial neural networks.
- **Topology:** Human brains have no “layers”. **Async:** The human brain works asynchronously, ANNs work synchronously.
- **Learning algorithm:** ANNs use gradient descent for learning. We don't know what human brains use
- **Power consumption:** Biological neural networks use very little power compared to artificial networks
- **Stages:** Biological networks usually never stop learning. ANNs first train then test.

1. Parameters (参数数量差异巨大)

Human brains have ~10,000,000 times synapses than ANNs.

✓ 生物神经网络参数量 (突触数)

人脑有大约：

$$10^{11} \text{ neurons} \times 10^3 \text{ synapses per neuron} = 10^{14} \text{ synapses}$$

也就是：

100 trillion parameters

✓ 人工神经网络参数量

- AlexNet: 60M
- GPT-2: 1.5B
- GPT-3: 175B
- GPT-4 (估计): ~1T

和人脑相比仍然小很多。

★ 生物大脑的复杂度完全碾压 ANN。

4. Learning algorithm (学习方式截然不同)

ANNs use gradient descent for learning.
We don't know what human brains use.

✓ ANN 的学习：

使用反向传播 (backprop) + 梯度下降 (gradient descent)：

$$W \leftarrow W - \eta \frac{\partial L}{\partial W}$$

✓ 生物学习机制：

我们并不知道确切算法，但可能包括：

- Hebbian Learning (Hebb 假说：fire together wire together)
- Synaptic plasticity (突触可塑性)
- Neurochemical signaling (神经递质调节)

★ 大脑不会做梯度下降！ANN 只是受到启发而不是模拟人脑。

2. Topology (拓扑结构不同)

Human brains have no “layers”.

✓ 人工神经网络 (ANN)

- 典型结构是：Layer 1 → Layer 2 → Layer 3 → Output
- 严格的层级式、前向结构

✓ 人脑结构

- 没有“层”的概念
- 连接极其复杂
- 全连接、环路、反馈无处不在
- 网络随机但高度组织化

★ 大脑远不像人工神经网络那样简单层级。

5. Power Consumption (能耗)

Biological neural networks use very little power compared to artificial networks.

✓ 生物大脑能耗

- 人脑只需要 20W
- 和一个灯泡差不多

✓ ANN 能耗

- GPT 模型训练一次要消耗百万瓦时
- 大模型训练需要大型数据中心
- GPU 集群能耗极高

★ 大脑是史上最节能的计算系统，ANN 的能效远远不如。

3. Async / Sync (同步 vs 异步)

The human brain works asynchronously, ANNs synchronously.

✓ 生物神经元

- 各神经元独立放电
- 没有统一时钟
- 完全异步 (asynchronous)

✓ 机器学习中的 ANN

- 一层一层严格同步计算
- 所有节点同时前向、同时反向
- GPU 上必须同步批量进行

★ 大脑是异步自组织系统，ANN 是严格同步的数学系统。

6. Stages (学习模式差异)

Biological networks never stop learning.
ANNs first train then test.

✓ ANN 的学习流程：

1. 训练 (train)
2. 停止训练
3. 测试 (test)

训练期间可更新权重，测试期间不可更新。

✓ 生物神经网络 (大脑)：

- 人类从出生到死亡都在不断修改突触
- 没有所谓“train”和“test”阶段的分离
- 体验、记忆、学习是持续发生的

★ 大脑是持续学习系统，ANN 是 offline 学习系统。